Jingze (Jacob) Ma

213-662-0232 | jingzema2001@gmail.com | Los Angeles, CA | Website | LinkedIn | No Sponsor Needed

EDUCATION

Yale University, New Haven, CT — May 2025
M.S. in Computer Science
GPA: 4.0/4.0
Selected coursework: Building Distributed Systems, Database System, Full Stack Web Programming, Linear Model

University of Southern California, Los Angeles, California — May 2024
B.S./B.A., Double Major: Computer Science (Games); Computational and Applied Mathematics
GPA: 3.94/4.00, Summa Cum Laude
Selected coursework: Game Engines, Machine Learning, Algorithms, Data Structures, Linear Algebra, Discrete Methods

SKILLS

Programming Languages: C++, C#, C, Java, Swift, Kotlin, Go, SQL, JavaScript, TypeScript, HTML/CSS, Python
Tools and Frameworks: React, Next.js, Node.js, SwiftUI, Spring Boot, PostgreSQL, MySQL, Redis, Kafka, AWS, Docker, Firebase, CMake

PROFESSIONAL EXPERIENCE

Software Engineer, **Smule, Inc.**, Remote — July 2026 – Present
Machine Learning Engineer Intern — *Apr 2026 – July 2026*

- Built the iOS client for Smule Sing's redesigned search in Swift and SwiftUI (MVVM on a state-machine workflow architecture), covering results, sort and filter, pagination, and analytics behind a server-side A/B test flag.
- Designed the Kotlin architecture for the Android version of the redesign, approved in architecture review, and am implementing it to reach feature parity with iOS.
- Designed bot protection for smule.com search and render endpoints using Cloudflare Turnstile, with server-side verification in Node.js and Redis-cached passes, to curb scraping and costly automated re-renders.
- Built a Kafka event sampler (Avro, Schema Registry) and evaluated ingestion options for ~240M events/day to supply training data for recommendation model fine-tuning; supported backend services across web and mobile.

Associate Software Engineer, **Vestmark, Inc.,** Wakefield, Massachusetts — June 2025 – Jan 2026

- Developed and enhanced Spring Boot REST APIs for VestmarkONE Advisor Trading, powering session-based workflows such as Rebalance-to-Model and Tax Harvest Buyback used by the Advisor UI.
- Implemented ad-hoc trading features, including compliance checks with drift-factor exceptions during enter-trade sessions, plus new endpoints and payload extensions for trading and account metrics.
- Improved API correctness by hardening validation and error handling across session and account endpoints.
- Ran API scale tests in Jenkins with JMeter and Selenium to verify scalability and reliability.

Software Engineer Intern, **eBaoTech International, Ltd**., Shanghai, China — June 2023 – Aug 2023

- Designed a programming language translation pipeline for the InsureMO platform by fine-tuning large language models (LLMs), including ChatGLM v2, with PyTorch and Hugging Face.
- Trained and deployed pet insurance premium prediction models on AWS SageMaker and S3.

Software Engineer Intern, **nPlace, Ltd.**, Shanghai, China — June 2022 – Aug 2022

- Built adaptive downsampling for a sensor data converter to stabilize refresh density, holding peak pixels per frame below 20k and the average below 10k without significant accuracy loss.
- Implemented and presented integer (ground truth) and floating-point versions of the 3D Bresenham line algorithm, including fixes to its error term.
- Contributed to the App Store release of Zaichang, a real-time 3D scanning and modeling app.

SELECTED PROJECT EXPERIENCE

Founder & Full Stack Engineer, **ROAMY Trip,** New Haven, Connecticut — Jan 2025 – Current

- Lead a team of 5 engineers building ROAMY Trip, a cross-platform social travel planning app inspired by Beli, with a React Native frontend and a Node.js/Express backend on PostgreSQL.
- Designed and implemented all frontend pages and the API integration for feed and rating features.
- Coordinate team operations, including task delegation and project management.

Gameplay and AI Engineer, **Lake Minnewaska (Steam)**, Los Angeles, California — Jun 2023 – May 2024

- Collaborated with a team of 30 to develop a fishing simulator game in Unity.
- Designed and implemented the fishing system, including casting and reeling mechanics built on state machines.
- Built game AI for fish behavior and NPC interactions using behavior trees and pathfinding.
- Developed key gameplay features, including the fishing journal, cutscenes, and UI.

Student Researcher, **Determine the Influence of a Movie Using Machine Learning**, Remote — May 2020 – Aug 2020

- Analyzed 30k reviews of 1k+ movies with LSTM sentiment analysis to identify key factors in a movie's influence.
- Published the paper at 2020 2nd International Conference on Machine Learning (Link)